JavaScript Array Traversal using `for` Loop

The `for` **loop** is the most commonly used loop for traversing (iterating through) arrays in JavaScript.

It allows you to access every element one by one using its **index**.

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What is a for Loop?

Definition

A `for` loop repeatedly executes a block of code until a specified condition becomes `false` .

When working with arrays, it is mainly used to access every element using its index.

Why Use a for Loop?

We use a `for` loop to:

- Traverse an array
- Print array elements

- Calculate totals
- Search data
- Update array values
- Process API data

Syntax

```
for(initialization; condition; update){

    // code

}
```

Array Syntax

```
for(let i = 0; i < array.length; i++){

    console.log(array[i]);

}
```

Understanding the Loop

```
const fruits = [

    "Apple",

    "Mango",

    "Orange"

];
```

Iteration	i	fruits[i]
1	0	Apple
2	1	Mango
3	2	Orange

When `i` becomes `3`, the condition

```
i < fruits.length
```

becomes

```
3 < 3
```

which is `false`, so the loop stops.

Example 1 (Print All Elements)

```
const fruits = [
    "Apple",
    "Mango",
    "Orange"
];
for(let i = 0; i < fruits.length; i++){
    console.log(fruits[i]);
}
```

Output

```
Apple
Mango
Orange
```

Example 2 (Numbers)

```
const numbers = [
    10,
    20,
    30,
    40
];
for(let i = 0; i < numbers.length; i++){
    console.log(numbers[i]);
}
```

Output

```
10
20
30
40
```

Example 3 (Using Index)

```
const colors = [
```

```
    "Red",
    "Green",
    "Blue"
];
for(let i = 0; i < colors.length; i++){
    console.log(i, colors[i]);
}
```

Output

```
0 Red
1 Green
2 Blue
```

Example 4 (Sum of Array)

```
const numbers = [
    10,
    20,
    30,
    40
];
let sum = 0;
for(let i = 0; i < numbers.length; i++){
    sum += numbers[i];
```

```
}

console.log(sum);
```

Output

```
100
```

Example 5 (Find Largest Number)

```
const numbers = [
    20,
    60,
    15,
    80,
    40
];
let largest = numbers[0];
for(let i = 1; i < numbers.length; i++){
    if(numbers[i] > largest){
        largest = numbers[i];
    }
}
console.log(largest);
```

Output

```
80
```

Example 6 (Print Even Numbers)

```
const numbers = [
    2,
    5,
    8,
    11,
    20
];
for(let i = 0; i < numbers.length; i++){
    if(numbers[i] % 2 === 0){
        console.log(numbers[i]);
    }
}
```

Output

```
2
8
20
```

Example 7 (Update Every Element)

```
const prices = [
    100,
    200,
    300
];
for(let i = 0; i < prices.length; i++){
    prices[i] = prices[i] + 50;
}
console.log(prices);
```

Output

```
[150,250,350]
```

Example 8 (Reverse Traversal)

```
const fruits = [
    "Apple",
    "Mango",
    "Orange"
];
for(let i = fruits.length - 1; i >= 0; i--){
```

```
    console.log(fruits[i]);

}
```

Output

```
Orange
Mango
Apple
```

Example 9 (Search an Element)

```
const students = [

    "Rahim",

    "Karim",

    "Rokon"

];

let found = false;

for(let i = 0; i < students.length; i++){

    if(students[i] === "Rokon"){

        found = true;

        break;

    }

}

console.log(found);
```

Output

```
true
```

Example 10 (Count Elements)

```
const fruits = [
    "Apple",
    "Mango",
    "Orange",
    "Banana"
];
let count = 0;
for(let i = 0; i < fruits.length; i++){
    count++;
}
console.log(count);
```

Output

```
4
```

Real Project Example 1

Shopping Cart

```
const cart = [
    "Laptop",
    "Mouse",
    "Keyboard"
];
for(let i = 0; i < cart.length; i++){
    console.log(cart[i]);
}
```

Real Project Example 2

Student Result

```
const marks = [
    80,
    75,
    95
];
let total = 0;
for(let i = 0; i < marks.length; i++){
    total += marks[i];
}
console.log(total);
```

Output

```
250
```

Real Project Example 3

Product Price

```
const prices = [
    100,
    200,
    300
];
for(let i = 0; i < prices.length; i++){
    console.log("Price:", prices[i]);
}
```

Real Project Example 4

Attendance List

```
const students = [
    "Rahim",
    "Karim",
    "Rokon"
```

```
];

for(let i = 0; i < students.length; i++){

    console.log("Present:", students[i]);

}
```

Best Practices

Use `array.length` instead of writing fixed numbers.

```
for(let i = 0; i < array.length; i++)
```

Use meaningful variable names.

```
students

products

prices

marks
```

Keep loop logic simple and readable.

Common Mistakes

Mistake 1

Using `<=`

Wrong

```
for(let i = 0; i <= array.length; i++)
```

Correct

```
for(let i = 0; i < array.length; i++)
```

Mistake 2

Forgetting `i++`

Wrong

```
for(let i = 0; i < array.length;){

}
```

This creates an infinite loop.

Mistake 3

Starting from the wrong index.

Wrong

```
for(let i = 1; i < array.length; i++)
```

This skips the first element.

Mistake 4

Accessing an invalid index.

```
array[array.length]
```

returns

```
undefined
```

Interview Questions

What is the purpose of a `for` loop?

It repeats a block of code until a condition becomes false.

Why is `array.length` used?

To avoid going beyond the last element.

Which index does an array start from?

```
0
```

How do you print an array in reverse order?

```
for(let i = array.length - 1; i >= 0; i--){
    console.log(array[i]);
}
```

How do you find the sum of all elements?

Use a variable and add each element inside the loop.

Summary Table

Feature	Description
Loop	`for`
Purpose	Traverse an array
Starts From	`0`
Ends At	`array.length - 1`
Access Element	`array[i]`
Changes Array	Optional
Common Uses	Print, Search, Sum, Update, Traverse

Final Notes

The `for` **loop** is the foundation of array traversal in JavaScript. Before learning advanced methods like `for...of`, `map()`, `filter()`, and `reduce()`, you should be comfortable using the `for` loop because it gives you complete control over the array index and iteration process.